



UTM
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BACHELOR OF COMPUTER SCIENCE (DATA ENGINEERING) WITH HONS.

FACULTY OF COMPUTING

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SECP3213 - BUSINESS INTELLIGENCE

INDUSTRY TALK REPORT

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1.0 Introduction

On 20th May 2026, the Faculty of Computing at Universiti Teknologi Malaysia (UTM) Johor Bahru welcomed Mr. Faiz Fablillah as the guest speaker for an industry lecture titled "Bridging Data & Decisions: The Role of Data Governance and Data Management in Modern Business Intelligence." Mr. Faiz is a Senior Practitioner in Data & AI Governance and holds a Certified Data Management Professional (CDMP) certification under DAMA International. With years of hands-on experience navigating real enterprise data challenges, he was well-positioned to speak on a topic that many organisations continue to struggle with, making sense of their data in a way that actually drives decisions.

The talk was aimed at showing how data governance and proper data management are not just technical checkboxes, but the backbone of any meaningful Business Intelligence effort. Mr. Faiz made it clear early on that having vast amounts of data means very little if that data is poorly managed, inconsistent, or ungoverned, in his words, it becomes more of a liability than an asset. For students in SECP3213 Business Intelligence, the session was particularly eye-opening because it put a real-world face on concepts we had only seen in textbooks and lectures. Topics like data quality, compliance, and governance frameworks suddenly felt a lot more tangible when discussed through the lens of actual industry experience. Overall, the talk served as a strong reminder that the foundation of good Business Intelligence is not the fanciest tool or the most complex algorithm, it is the quality and trustworthiness of the data sitting underneath it all.

2.0 Key Takeaways

2.1 The Role and Responsibilities of an Analytics Engineer

The speaker emphasized that while textbooks cleanly categorize data roles, real-world data is highly scattered, and every company handles it differently. Consequently, a modern Analytics Engineer is rarely hired just to build dashboards. They must possess end-to-end understanding, from data sources and engineering requirements to stakeholder needs. Because data issues are frequent, communication is the primary way of work. Resolving discrepancies and aligning technical requirements require constant cross-departmental collaboration.

2.2 Bridging the Gap Between Raw Data and Business Decisions

Analytics transforms raw operational data into strategic business decisions. The speaker highlighted that this requires balancing two "sibling" concepts:

1. **Data Governance:** The strategic framework (rules, laws, and ownership) that ensures safety and compliance. Without governance, there is no trusted data.
2. **Data Management:** The operational execution, development, and control of data activities.

To bridge the gap, data policies must reflect practical data management and align with business goals. Everything is interconnected: Data Quality (DQ) is ineffective without Business Intelligence (BI) to visualize it, and BI is useless without reliable DQ.

2.3 Core Data Practices and Technologies

While formal frameworks outline eleven distinct pillars of data management, the speaker emphasized four foundational pillars as the most critical for modern organizations looking to establish a trusted "single source of truth":

1. **Metadata & Data Cataloging:** Metadata provides the data's description and ownership details. Cataloging creates awareness that a dataset exists, boosting data usage across the company.
2. **Data Lineage:** Tracing data from origin to its final form. Though difficult to implement, it is vital for tracking data health.
3. **Data Security & Access Control:** Classifying data to restrict visibility. For example, a confidential dashboard built for the CEO must be locked down so only the CEO can access it.
4. **Master Data Management (MDM):** Managing shared enterprise data by combining different data types:
 - *Master Data:* Core records shared across departments (maintains only the latest record).
 - *Reference Data:* Static data that rarely changes (e.g., names or addresses).
 - *Transactional Data:* Historical records of business activities.
 - *Analytical Datasets:* The powerful combination of Master and Transactional data used for insights.

2.4 Real-World Realities and Future Trends

The data landscape has shifted drastically. In the past, manual operations were slow, leaving large amounts of data completely ungoverned. Today, the industry is aggressively moving toward **utilizing AI**, particularly to automate data quality checks and eliminate manual bottlenecks. Ultimately, organizations must realize that these data activities are entirely related and must execute multiple interconnected pieces to achieve the target business outcome.

3.0 Reflection and critical insights

The most valuable takeaway from this talk was how the real BI pipeline actually works, which exposed the complexity behind every dashboard. Several problems that are usually faced before the final visualisation are such as integration challenges, data quality issues, security concerns, and compliance requirements. This has reframed our understanding of BI in visualisation tools into a full engineering and governance discipline. The DAMA-DMBOK framework was equally insightful, by showing data governance at the center of eleven knowledge areas. It shows that being a competent BI professional goes far beyond building reports. This connects well to course topics like data warehousing and ETL, though the talk added a compliance dimension that is rarely emphasised in class, Malaysia PDPA 2010, which made the legal consequences of poor data governance feel real and locally relevant. The slides about AI in Data Quality also stood out as the most forward looking topics because it highlighted how automated anomaly detection and predictive quality checks are replacing manual data cleaning. This has suggested future practitioners need to govern AI-assisted pipelines, not just build them.

Overall, analytics engineering is a critical role in today's business landscape because it bridges the raw data and reliable decisions. The talk has inspired further questions that are worth exploring more in the future such as how do organizations implement PDPA compliance within BI pipeline at scale? And as AI takes over data quality tasks, what new governance risks could be? These are areas we intend to explore as we advance in this field.

4.0 Conclusion

In the industry talk, Mr. Faiz Fablillah shared his insights on Business Intelligence and highlighted that the key to driving business decisions is the right data, which is safeguarded by robust governance and PDPA 2010 adherence. We understood that BI can only be successful

when there's collaboration among different departments, accurate data lineage, and data quality management with a proactive approach to ensure a single source of truth.

This lecture was a reminder to us as future data professionals to build our technical expertise beyond data, to data cataloging, Master Data Management (MDM), and managing AI-assisted pipelines. Today, it is understood that responsibilities don't end with the creation of dashboards, they must be data integrity stewards. In this session we have been prepared industry ready mindset for designing secure, accountable and business centric BI solutions.